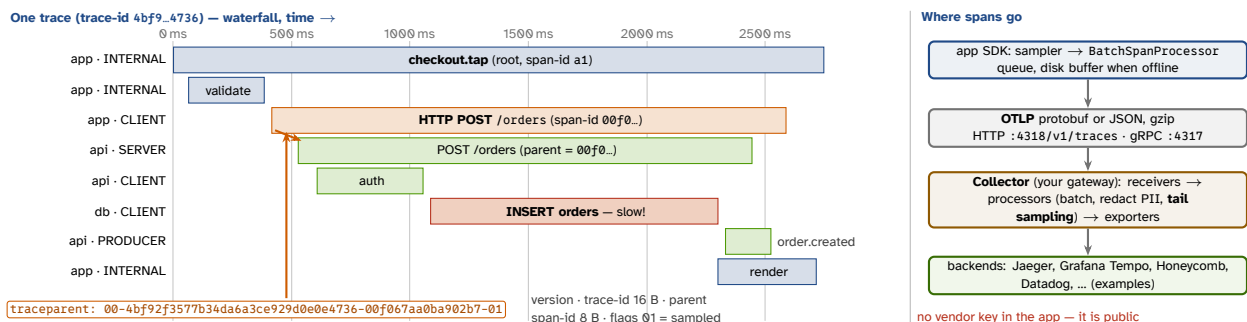


In one line: **OpenTelemetry** (OTel) is the vendor-neutral standard for telemetry: an **API** you instrument against, an **SDK** that samples, batches and exports, the **OTLP** wire protocol and a **Collector**. A **trace** is a tree of **spans** sharing one trace-id; the app starts it at the user's tap, and the W3C traceparent header carries it into every backend service — so one timeline shows where the 3 seconds went.



The model

- **Span** = name, trace-id, span-id, parent, **kind** (INTERNAL, CLIENT, SERVER, PRODUCER, CONSUMER), start/end, **attributes**, **events** (timestamped points), **links** (to other traces), **status** (Unset / Ok / Error). A **trace** = all spans with one trace-id.
- **Metrics**: counters, histograms, gauges — aggregated, cheap, never sampled. **Logs**: records stamped with trace/span id for correlation.
- **Resource** = who emitted it (service.name, version, OS, device). **Semantic conventions** fix names: `http.request.method`, `url.full`, `http.response.status_code`, `session.id`.
- **API vs SDK**: libraries call only the API (no-op without an SDK); the app configures the SDK: provider, sampler, processors, exporter.
- **Context** holds the active span; a **propagator** injects it into requests and the server extracts it. **Baggage** (baggage header): never PII.

Example — a manual root span (Swift)

```
import OpenTelemetryApi
let tracer = OpenTelemetry.instance.tracerProvider
    .get(instrumentationName: "com.example.checkout",
        instrumentationVersion: "1.4.0")
let span = tracer.spanBuilder(spanName: "checkout.tap")
    .setSpanKind(spanKind: .internal)
    .setActive(true) // URLSession spans become children
    .startSpan()
span.setAttribute(key: "cart.items", value: cart.count) // no PII
do { try await api.placeOrder(cart) // CLIENT span + traceparent
    span.status = .ok }
catch { span.status = .error(description: "(typeOf: error)") }
span.end() // on EVERY path
```

SDKs — status on opentelemetry.io (Sep 2026)

Swift	opentelemetry-swift: traces stable , metrics + logs in development. URLSession instrumentation (spans + header injection, shouldInjectTracingHeaders), SDKResourceExtension, signpost integration
Android	opentelemetry-android RUM agent, alpha , on the Java SDK: lifecycle, ANR, crashes, slow/frozen frames, OkHttp, startup, sessions (API 23+)
Web (JS)	traces + metrics stable, logs in development; fetch/XHR/document-load instrumentations
React Native	no official SDK — JS API + native SDKs, or a vendor SDK

Sampling — who decides what you pay for

- **Head** sampling: decided at the root, before the outcome is known — `TraceIdRatioBased(0.1)` wrapped in `ParentBased`, so every child (and the backend) follows the 01 flag. Cheap, consistent, but blind to errors.

- **Tail** sampling: in the Collector after the trace completes — keep all errors, all traces > 2 s, 5 % of the rest. Needs every span of a trace on the same Collector (route by trace-id) and memory to wait.
- Mobile: head-sample on device (millions of devices, metered radios); errors and crashes via an always-on path.

Export from a phone

- BatchSpanProcessor (queue + size/time flush); SimpleSpanProcessor only for debugging. OTLP retries 429/502/503/504 with backoff; honour `partial_success`.
- Flush when the app backgrounds (short background task), buffer to disk offline, cap the queue (drop oldest, count drops), gzip, batch to save radio wake-ups.
- Send to **your** Collector/gateway: it authenticates the app, redacts, samples, and lets you switch vendors by config.

What to instrument in an app

app start	process start → first frame (cold/warm as an attribute)
screen load	appear → content ready; name the screen
network	auto CLIENT spans + traceparent to your hosts only
user flows	tap → result as the root (checkout, login, search)
local work	DB queries, decoding, image processing, long main-thread work
state	app version, network type, session.id, feature-flag arms

Privacy

No PII in names, attributes, events, baggage: strip URL queries and ids, pseudonymise user ids. Honour consent; redact again in the Collector; declare it in the privacy manifest.

Traps

- ! The backend starts a new trace — propagator not configured, or a proxy/CDN strips traceparent. Check the header on the wire.
- ! Device clocks drift: a child "starts before" its parent. Trust durations, not cross-device offsets.
- ! User ids or raw URLs as *metric* attributes — cardinality explosion; use route templates (`/orders/{id}`).
- ! A span not ended on the error path is never exported (and leaks).
- ! Injecting traceparent into third-party hosts: leaks ids, can break CORS/signing.

Remember

Spans say what and how long, context says whose, traceparent carries it across, sampling decides what you pay for, the Collector decides where it goes.